# Semi-supervised Medical Image Segmentation Based on Mutual Consistency Learning

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**Abstract**

In this work, we present a novel mutual consistency network (MC-Net+) designed to effectively harness unlabeled data for semi-supervised medical image segmentation. The development of MC-Net+ is driven by a key observation: deep models trained on sparsely annotated data tend to produce highly uncertain and error-prone predictions in anatomically ambiguous regions (*e.g.,* adhesive boundaries or slender branch structures). Capitalizing on these difficult samples can substantially enhance the training efficacy of semi-supervised segmentation models. Accordingly, MC-Net+ incorporates two core innovations. First, the architecture comprises a shared encoder coupled with multiple subtly distinct decoders (*i.e.,* employing different up-sampling strategies). The statistical discrepancy among the outputs of these decoders serves as a proxy for model uncertainty, thereby highlighting unlabeled challenging regions. Second, we introduce a mutual consistency constraint that enforces agreement between one decoder's probability output and the soft pseudo labels generated by the remaining decoders. This mechanism progressively reduces the output discrepancy (*i.e.,* the model uncertainty) during training and compels the network to yield consistent predictions in difficult regions, thereby serving as an effective regularizer. We benchmark MC-Net+ against five state-of-the-art semi-supervised methods on three publicly available medical datasets. Extensive experiments under two standard semi-supervised protocols demonstrate the superiority of our approach, establishing a new state of the art for semi-supervised medical image segmentation.

*Keywords:* Mutual consistency, soft pseudo label, semi-supervised learning,

## 1. Introduction

Accurate medical image segmentation constitutes a cornerstone of effective computer-aided diagnosis (CAD) systems. Reliable segmentation results enable the quantitative analysis of morphological attributes of organs and tissues, providing clinicians with a valuable basis for disease diagnosis. Moreover, a robust segmentation model facilitates the localization of specific anatomical structures, which is essential for early disease screening and precise clinical assessment (Hesamian et al., 2019).

The past few years have witnessed remarkable advances in deep learning–based medical image segmentation. Nevertheless, these methods continue to exhibit suboptimal performance on a variety of clinical tasks. This limitation is largely attributable to overfitting caused by insufficient training data, as most medical image segmentation datasets remain relatively small in scale. The acquisition of densely annotated medical images is inherently expensive: pixel- or voxel-level annotation of medical images (*e.g.*, volumetric CT or MRI scans) demands specialized expertise, sustained concentration, and considerable time. Consequently, semi-supervised approaches that leverage unlabeled medical data have emerged as a critically important direction for improving segmentation performance and have garnered increasing research interest.

Existing semi-supervised methods can be broadly categorized into two families. The first comprises consistency-based models (Yu et al., 2019; Luo et al., 2021a,b), which rest on the smoothness assumption—that small perturbations applied to the input should not lead to substantial changes in the corresponding outputs (Laine and Aila, 2016). The second family encompasses entropy-minimization techniques (Lee et al., 2013; Rizve et al., 2021; Pham et al., 2021), which build upon the cluster assumption—namely, that samples belonging to the same class should form compact clusters with low entropy. However, the majority of existing approaches fail to adequately exploit the varying learning difficulties (Soviany et al., 2021) present in unlabeled data. Given that deep models naturally produce pixel-/voxel-level uncertainty estimates alongside their segmentation outputs, we argue that leveraging such uncertainty information can lead to more effective utilization of unlabeled data, ultimately advancing semi-supervised medical image segmentation.

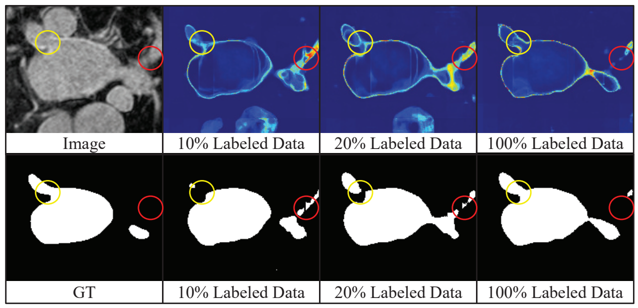


Figure 1: Three exemplar uncertainty maps and the corresponding segmentation results produced by a fully supervised V-Net model trained with 10%, 20%, and all labeled data on the LA dataset.

Rather than adopting a curriculum learning strategy (Soviany et al., 2021), our central idea is to directly exploit the challenging unlabeled regions to facilitate model training. We use Fig. 1 to further illustrate this motivation. Specifically, the figure presents three uncertainty maps and their corresponding segmentation results on the left atrium (LA) dataset, obtained from three fully supervised V-Net models trained with 10%, 20%, and all labeled data, respectively. Each uncertainty map was generated via the Monte Carlo Dropout (MC-Dropout) technique following (Yu et al., 2019). Two key observations emerge from Fig. 1: *(1) The most uncertain predictions are predominantly concentrated in anatomically challenging regions (e.g., thin branch junctions, as highlighted by the yellow and red circles in Fig. 1).* In contrast, regions devoid of complex textures or variable appearance are segmented with high accuracy. Put differently, as more labeled data become available for training, the V-Net model primarily refines its predictions in a small number of difficult areas. *(2) As the amount of labeled training data increases, the model tends to produce* *less ambiguous predictions.* This suggests that the generalization capability of deep models is closely tied to their predictive uncertainty. These observations motivate us to explore model uncertainty as a means of guiding the model toward better generalization in challenging regions, a direction that aligns with the concurrent work of (Qiao and Peng, 2021).

Building upon these insights, we propose a novel mutual consistency network (MC-Net+) for semi-supervised medical image segmentation, with the aim of directing greater attention toward unlabeled challenging regions via estimated model uncertainty. As illustrated in Fig. 2, MC-Net+ consists of a single shared encoder and multiple subtly distinct decoders. The statistical discrepancy among the outputs of these decoders is used to capture pixel-/voxel-level uncertainty, thereby identifying hard regions. Additionally, we employ a sharpening function to convert probability outputs into soft pseudo labels and design a novel mutual consistency training scheme that enforces agreement between one decoder's probability output and the soft pseudo labels generated by the other decoders. This mechanism progressively minimizes the output discrepancy across decoders during training and establishes an end-to-end learning framework, enabling the model to extract more informative features from unlabeled challenging regions.

In summary, the contributions of this paper are three-fold.

- We propose the MC-Net+ model for semi-supervised segmentation, grounded in the key insight that enforcing consistent, low-entropy predictions in challenging regions enables more effective exploitation of unlabeled data and yields improved segmentation performance.
- We design a novel mutual consistency scheme that unifies both consistency and entropy-minimization constraints, thereby encouraging the model to learn a more generalizable feature representation.
- Extensive experiments demonstrate that MC-Net+ outperforms five recent methods and establishes a new state of the art (SOTA) for semi-supervised medical image segmentation.

A preliminary version of this work was presented at MICCAI 2021 (Wu et al., 2021), which encouraged mutual consistency between two slightly different decoders. The present paper substantially extends the conference version with the following contributions: (1) incorporating an additional decoder based on nearest-neighbor interpolation into the original MC-Net model, thereby further enhancing intra-model diversity; (2) conducting experiments on the additional Pancreas-CT and ACDC datasets to demonstrate the general effectiveness of our approach across diverse semi-supervised medical image segmentation tasks; (3) reimplementing five recent methods within a unified experimental environment and reporting their results for fair comparisons; (4) supplementing the discussion with hyper-parameter analyses; (5) adopting the original encoder–decoder architecture for inference, thus introducing no additional computational overhead at test time.

## 2. Related Work

### *2.1. Semi-supervised Learning*

Semi-supervised learning (SSL) has been extensively studied across a broad range of computer vision tasks. Among consistency-based approaches, various data augmentation strategies (Zhang et al., 2017; Xie et al., 2019; Wu et al., 2022) have been employed to generate perturbed versions of the training data. For instance, Ouali et al. (2020) applied multiple data augmentation operations to perturb intermediate feature maps and constrained the model to produce invariant segmentation outputs. Wang et al. (2022) leveraged semantic directions in the feature space for semantic data augmentation and subsequently imposed consistency constraints for SSL. Sohn et al. (2020) exploited the consistency between weakly and strongly perturbed views of the same training sample to facilitate learning. Consistency at the model level has also been explored in the mean-teacher framework through the use of an exponential moving average (EMA) update strategy (Tarvainen and Valpola, 2017). In addition, adversarial training (Miyato et al., 2018; Mittal et al., 2019; You et al., 2022b) has been adopted to impose stronger consistency constraints during training.

Entropy-minimization-based models constitute another effective line of work for boosting semi-supervised learning. For example, Kalluri et al. (2019) introduced an entropy minimization module that encourages the model to produce low-entropy predictions on unlabeled samples. Pseudo label learning (Lee et al., 2013; Chen et al., 2021) typically employs a sharpening function or a fixed threshold to convert probability maps into pseudo labels; under the supervision of these pseudo labels, the model learns to generate low-entropy outputs. Rizve et al. (2021) combined probability and uncertainty thresholds to select the most reliable pseudo labels for SSL. Pham et al. (2021) incorporated a meta-learning scheme into pseudo label learning to further enhance performance.

It is now widely recognized that both consistency and entropy-minimization constraints can improve the discriminative power of semi-supervised models. In this work, we integrate both techniques into our MC-Net+ model to achieve accurate semi-supervised medical image segmentation.

### *2.2. Semi-supervised Medical Image Segmentation*

A number of recent methods have been proposed to address semi-supervised medical image segmentation. Yu et al. (2019) introduced an uncertainty-aware mean-teacher model for semi-supervised left atrium segmentation. Li et al. (2020b) further imposed shape constraints by incorporating signed distance maps (SDMs) (Ma et al., 2020) to improve segmentation quality. Luo et al. (2021a) investigated the relationship between medical image segmentation and organ shape regression. They also explored a semi-supervised framework that enforces multi-scale consistency for gross target volume segmentation (Luo et al., 2021b). Xia et al. (2020b,a) adopted a multi-view co-training strategy to perform ensemble learning for 3D medical image segmentation. Xie et al. (2020) leveraged attention mechanisms to capture pairwise relationships between labeled and unlabeled data, thereby alleviating overfitting caused by limited annotations.

Although these approaches have demonstrated promising results, they largely overlook or underestimate the impact of unlabeled challenging regions during training. We hypothesize that the performance of semi-supervised segmentation can be further improved by more effectively modeling these difficult regions, even in the absence of corresponding ground-truth labels. It is worth noting that the CPS model (Chen et al., 2021) recently introduced a cycled-consistency framework that bears similarity to ours for semi-supervised image segmentation. However, their model employs identical architectures initialized with different parameters and perturbs the input images with different noise. In contrast, our approach is motivated by the uncertainty analysis in Fig. 1 and adopts a shared encoder with multiple subtly distinct decoders. Section *6.1* provides further evidence that employing different up-sampling strategies yields superior segmentation results.

### *2.3. Multi-task Learning*

Another promising avenue for improving the generalization of deep models lies in learning cross-task feature representations or performing unsupervised pre-training (You et al., 2021). Contrastive learning–based methods (Chaitanya et al., 2020) can carry out self-supervised training to mitigate overfitting. For instance, You et al. (2022a) employed global and local contrastive learning to extract more generalizable features from unlabeled data, achieving substantial performance gains in semi-supervised medical image segmentation. Moreover, proxy or auxiliary tasks can be constructed to explicitly regularize model training (Zhu et al., 2020; You et al., 2020). In particular, shape or boundary constraints have been utilized for anatomical refinement in medical image segmentation (Ma et al., 2020; Murugesan et al., 2019). Auxiliary losses (*e.g.,* for image reconstruction) can also help the model learn more generalizable and informative features (Wang et al., 2021; Castillo-Navarro et al., 2020). You et al. (2022c) combined knowledge distillation with multi-task learning to effectively leverage unlabeled data, yielding favorable semi-supervised segmentation performance.

In contrast to these successful methods, our MC-Net+ model does not require the design of task-specific auxiliary objectives and focuses solely on the original segmentation task during training. At the same time, our method can be readily combined with multi-task learning frameworks to further boost semi-supervised medical image segmentation performance.

### *2.4. Uncertainty Estimation*

Uncertainty analysis has attracted considerable attention in both machine learning and computer vision (Abdar et al., 2021; Jungo and Reyes, 2019). Beyond obtaining accurate predictions, it is desirable to also quantify the confidence associated with those predictions. Aleatoric uncertainty arises from annotation noise, whereas epistemic uncertainty reflects the limitations of the model itself (Kendall and Gal, 2017). In the semi-supervised setting, our discussion is confined to epistemic uncertainty, which can in principle be reduced by providing additional training data.

Several methods have been proposed for estimating uncertainty. Jin et al. (2019) employed a variational U-Net (Esser et al., 2018) to represent model uncertainty. Epistemic uncertainty can also be quantified through model ensembling (Lakshminarayanan et al., 2016), which computes the statistical discrepancy among the outputs of multiple independently trained models. However, this strategy incurs substantial computational overhead. To address this limitation, the MC-Dropout method was introduced in the Bayesian modeling framework to approximate model uncertainty in a more efficient manner (Gal and Ghahramani, 2016). Specifically, the dropout operation effectively samples multiple sub-models from a single network, and the statistical discrepancy among their outputs serves as an estimate of model uncertainty, thereby obviating the need to train multiple models independently. In this work, inspired by (Zheng and Yang, 2021), our model pre-defines multiple sub-models prior to training, enabling the estimation of epistemic uncertainty in a single forward pass.

## 3. Method

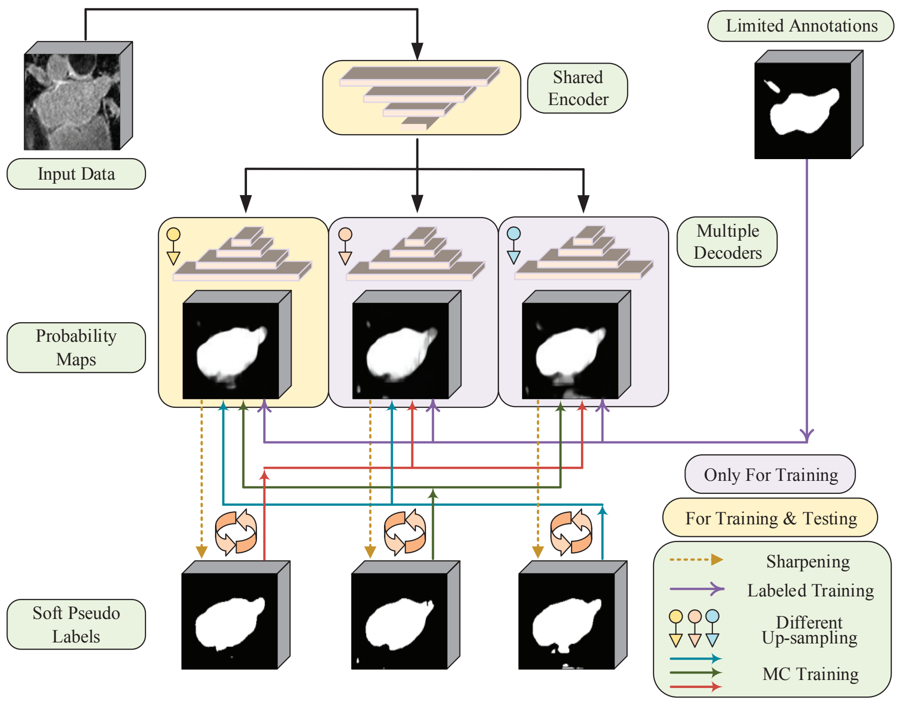


Figure 2: Overview of the proposed MC-Net+ model. A mutual consistency constraint is enforced between one decoder's probability output and the soft pseudo labels generated by the remaining decoders. Note that three subtly distinct decoders are employed in this work.

Before detailing the proposed model, we first formalize the semi-supervised segmentation problem and introduce the requisite notation. Let $x \in X$ denote an input image and $p(y_{pred}|x;\theta)$ the resulting probability map, where $\theta$ parameterizes a backbone network $f_\theta$. We use $y_l \in Y_l$ to denote the ground-truth segmentation annotations. The labeled and unlabeled datasets are defined as $\mathbb{D}_{\mathbb{L}} = \{x_l^i, y_l^i | i = 1, ..., N_l\}$ and $\mathbb{D}_{\mathbb{U}} = \{x_u^i | i = 1, ..., N_u\}$, respectively.

### 3.1. Model Architecture

The proposed MC-Net+ model is designed to exploit unlabeled ambiguous regions during training, where ambiguity is indicated by the model's epistemic uncertainty. Formally, the standard uncertainty estimation process can be expressed as follows:

$$\begin{aligned} &f_{\theta_{sub}} = Sampling(f_{\theta}) \\ &\mu_x = D[p(y_{pred}|x;\theta_{sub}^1), ..., p(y_{pred}|x;\theta_{sub}^n)] \end{aligned} \tag{1}$$

where $D$ computes the statistical discrepancy among $n$ outputs and $\mu_x$ represents the pixel-/voxel-level uncertainty. In the original MC-Dropout framework, the dropout operation samples $n$ sub-models $f_{\theta_{sub}}$ over $n$ separate forward passes.

To circumvent the computational burden of running multiple forward passes (more than eight in (Yu et al., 2019; Rizve et al., 2021)), we design MC-Net+ with a single encoder and $n$ subtly distinct decoders, as shown in Fig. 2. With a shared encoder $f_{\theta_e}$, we pre-define $n$ sub-models before uncertainty estimation. The uncertainty $\mu_x$ for an input $x$ then becomes:

$$\begin{aligned} &f_{\theta_{sub}^i} = f_{\theta_e} \boxplus f_{\theta_d^i},\ i \in 1, ..., n \\ &\mu_x = D[p(y_{pred}|x;\theta_{sub}^1), ..., p(y_{pred}|x;\theta_{sub}^n)] \end{aligned} \tag{2}$$

where $\boxplus$ signifies that a sub-model $f_{\theta_{sub}^i}$ is composed of the shared encoder $f_{\theta_e}$ and one decoder $f_{\theta_d^i}$. Each sub-model $f_{\theta_{sub}^i}$ follows a standard encoder–decoder architecture such as V-Net or U-Net (Milletari et al., 2016; Ronneberger et al., 2015).

To strike a favorable balance between effectiveness and efficiency, we set $n = 3$ in this work (see Fig. 2). Specifically, we employ a transposed convolutional layer, a bilinear interpolation layer, and a nearest-neighbor interpolation layer to construct three subtly distinct sub-models, thereby increasing intra-model diversity. This design enables more accurate approximation of model uncertainty and consequently yields improved semi-supervised segmentation performance. Further analysis of these design choices is provided in Section *6.2*.

### *3.2. Training via Mutual Consistency Constraints*

Building upon the architecture described above, the discrepancy among the outputs of multiple sub-models serves as a proxy for model uncertainty, indicating the locations of challenging regions. Recognizing that both consistency and entropy-minimization constraints facilitate effective utilization of unlabeled data, we propose a novel mutual consistency training strategy that integrates both forms of regularization.

Specifically, we first apply a sharpening function (Xie et al., 2019) to convert an output probability map $p(y_{pred}|x;\theta)$ into a soft pseudo label $p^*(y^*_{pred}|x;\theta)$:

$$p^*(y^*_{pred}|x;\theta) = \frac{p(y_{pred}|x;\theta)^{1/T}}{p(y_{pred}|x;\theta)^{1/T} + (1 - p(y_{pred}|x;\theta))^{1/T}} \tag{3}$$

where $T$ is a temperature hyper-parameter that governs the degree of sharpening. An appropriate choice of $T$ ensures that the entropy-minimization constraint regularizes the model effectively without introducing excessive noise that would hinder training. A detailed discussion of this parameter is provided in Section *6.3*.

We then perform mutual learning (Zhang et al., 2018) between one decoder's probability output and the soft pseudo labels generated by the other decoders. This mechanism reduces the discrepancy among the $n$ outputs, guiding the model toward consistent predictions in highly uncertain regions. The advantages of this design are three-fold: (1) consistency constraints are enforced by encouraging invariant outputs across all sub-models; (2) under the supervision of soft pseudo labels, the model is driven to produce low-entropy predictions, thereby satisfying the entropy-minimization constraint; (3) MC-Net+ can be trained end-to-end without requiring multiple forward passes.

Finally, the overall training objective combines a supervised loss and a mutual consistency loss as follows:

$$L_{mc} = \sum_{i,j=1\ \&\ i\neq j}^{n} D[p^*(y^*_{pred}|x;\theta^i_{sub}), p(y_{pred}|x;\theta^j_{sub})] \tag{4}$$

$$Loss = \lambda \times \sum_{i=1}^{n} L_{seg}(p(y_{pred}|x_l;\theta^i_{sub}), y_l) + \beta \times L_{mc} \tag{5}$$

where $L_{seg}$ denotes the standard Dice loss for segmentation, and $D$ is the Mean Squared Error (MSE) loss computed between paired inputs $p^*(y^*_{pred}|x;\theta^i_{sub})$ and $p(y_{pred}|x;\theta^j_{sub})$. The coefficients $\lambda$ and $\beta$ balance the supervised loss $L_{seg}$ and the mutual consistency loss $L_{mc}$, respectively. Note that $L_{mc}$ is applied to both the labeled set $\mathbb{D}_{\mathbb{L}}$ and the unlabeled set $\mathbb{D}_{\mathbb{U}}$.

## 4. Experiment

### *4.1. Datasets*

We evaluated the proposed MC-Net+ model on the LA, Pancreas-CT, and ACDC datasets. The LA dataset (Xiong et al., 2021), which serves as the benchmark for the 2018 Atrial Segmentation Challenge[1], comprises 100 gadolinium-enhanced MRI scans with an isotropic resolution of $0.625 \times 0.625 \times 0.625$ mm. Because the official test set does not include publicly available annotations, we adopted the same fixed split[2] used in prior work (Yu et al., 2019; Li et al., 2020b; Luo et al., 2021a), allocating 80 scans for training and the remaining 20 for validation. All methods are evaluated on this common validation set to ensure fair comparisons.

The Pancreas-CT dataset (Clark et al., 2013) consists of 82 contrast-enhanced abdominal CT scans collected from 53 male and 27 female subjects at the National Institutes of Health Clinical Center[3]. The scans were acquired on Philips and Siemens MDCT scanners at a fixed in-plane resolution of $512 \times 512$ pixels, with slice thicknesses ranging from 1.5 to 2.5 mm. Following the data split used in the DTC model (Luo et al., 2021a), we designated 62 scans for training and evaluated performance on the remaining 20. Voxel intensities were clipped to the range $[-125, 275]$ Hounsfield Units (HU) as in (Zhou et al., 2019), and the data were resampled to an isotropic resolution of $1.0 \times 1.0 \times 1.0$ mm.

The ACDC (Automated Cardiac Diagnosis Challenge) dataset (Bernard et al., 2018) was collected from clinical examinations performed at the University Hospital of Dijon[4] and contains cardiac MRI samples (multi-slice 2D

[1] http://atriaseg2018.cardiacatlas.org
[2] https://github.com/yulequan/UA-MT/tree/master/data
[3] https://wiki.cancerimagingarchive.net/display/Public/Pancreas-CT
[4] https://www.creatis.insa-lyon.fr/Challenge/acdc/databases.html

cine MRI) from 100 patients for training. Following the benchmark protocol of (Luo, 2020), we adopted a fixed patient-level split[5] in which the training, validation, and test sets comprise 70, 10, and 20 patients, respectively. Unlike the LA and Pancreas-CT datasets, which involve 3D binary segmentation, the ACDC task requires 2D multi-class segmentation. Accordingly, our 2D MC-Net+ model is designed to segment three cardiac structures: the myocardium, the left ventricle, and the right ventricle.

### *4.2. Implementing Details*

**3D Segmentation:** Following established protocols (Yu et al., 2019; Li et al., 2020b; Luo et al., 2021a), we first cropped the 3D volumes around the ground-truth annotations with enlarged margins of $[10 \sim 20, 10 \sim 20, 5 \sim 10]$ voxels for the LA dataset and $[25, 25, 0]$ voxels for the Pancreas-CT dataset. The cropped volumes were then normalized to zero mean and unit variance. During training, we randomly extracted 3D patches of size $112 \times 112 \times 80$ from LA and $96 \times 96 \times 96$ from Pancreas-CT.

For data augmentation on the LA dataset, we applied random 2D rotation and horizontal flip. On both datasets, the batch size was set to 4, with each batch comprising two labeled and two unlabeled patches. The 3D backbone was a V-Net that uses trilinear interpolation for feature map upsampling. The 3D MC-Net+ model was trained for 15,000 iterations. At test time, a sliding window of size $112 \times 112 \times 80$ (LA) or $96 \times 96 \times 96$ (Pancreas-CT) was applied with a fixed stride of $18 \times 18 \times 4$ or $16 \times 16 \times 16$, respectively. The patch-based predictions were then recomposed to yield the final volumetric segmentation.

**2D Segmentation:** On the ACDC dataset, we similarly normalized the images to zero mean and unit variance. Random rotation and flip were applied for data augmentation. 2D patches of size $256 \times 256$ were randomly cropped, and the batch size was set to 24 (12 labeled and 12 unlabeled samples per batch). During inference, the scans were resized to $256 \times 256$ as input and

[5]https://github.com/HiLab-git/SSL4MIS/tree/master/data/ACDC

the predictions were subsequently restored to the original resolution. Our 2D MC-Net+ employed U-Net as the backbone, utilizing bilinear interpolation for feature map upsampling. The 2D model was trained for 30,000 iterations. All settings on the ACDC dataset followed the public benchmark (Luo, 2020) to ensure fair comparisons.

Across all datasets, training was performed using the SGD optimizer with an initial learning rate of $10^{-2}$ and a weight decay of $10^{-4}$. The sharpening temperature $T$ was set to 0.1. The loss weight $\lambda$ was set to 1 for 2D segmentation and 0.5 for 3D tasks. The weight $\beta$ was governed by a time-dependent Gaussian warm-up function (Laine and Aila, 2016), consistent with prior methods (Yu et al., 2019; Li et al., 2020b; Luo et al., 2021a; Wu et al., 2021). We evaluated two standard semi-supervised settings—training with 10% and 20% labeled data—following (Yu et al., 2019; Li et al., 2020b; Luo et al., 2021a). All compared methods were reimplemented and evaluated within an identical experimental environment (Hardware: Intel(R) Xeon(R) Gold 6150 CPU @ 2.70GHz, NVIDIA Tesla V100 GPU; Software: PyTorch 1.8.0, CUDA 11.2, Python 3.8.10; Random Seed: 1337). Following (Yu et al., 2019; Li et al., 2020b; Luo et al., 2021a; Wu et al., 2021), four metrics were adopted for quantitative evaluation: Dice, Jaccard, Average Surface Distance (ASD), and 95% Hausdorff Distance (95HD).

## 5. Result

### *5.1. Performance on the LA Dataset*

Fig. 3 presents representative segmentation results for two samples from the LA dataset, shown in both 2D and 3D views. The results were produced by five recent methods and our approach, arranged from left to right. As can be observed, MC-Net+ generates a more anatomically complete left atrium compared with the other SOTA methods. Importantly, no morphological post-processing (*e.g.,* largest connected component selection, as employed in (Li et al., 2020b)) was applied to refine the outputs. Our model inherently eliminates

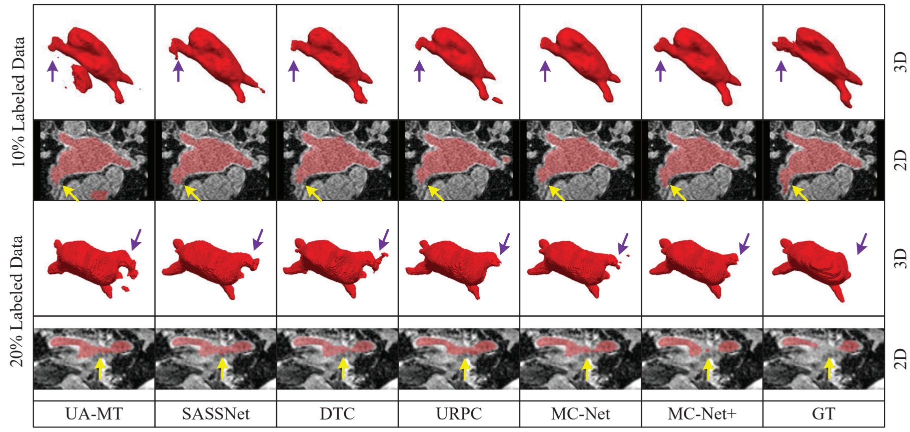


Figure 3: Exemplar segmentation results in 2D and 3D views on the LA dataset, produced by UA-MT (Yu et al., 2019), SASSNet (Li et al., 2020b), DTC (Luo et al., 2021a), URPC (Luo et al., 2021a), MC-Net (Wu et al., 2021), and our MC-Net+ model, with the corresponding ground truth shown on the right.

Table 1: Comparisons with five state-of-the-art methods on the LA dataset. Model complexities—the number of parameters (Para.) and multiply-accumulate operations (MACs)—are measured during inference.

| Method | # Scans used | | Metrics | | | | Complexity | |
|---|---|---|---|---|---|---|---|---|
| | Labeled | Unlabeled | Dice(%)↑ | Jaccard(%)↑ | 95HD(voxel)↓ | ASD(voxel)↓ | Para.(M) | MACs(G) |
| V-Net | 8(10%) | 0 | 78.57 | 66.96 | 21.20 | 6.07 | 9.44 | 47.02 |
| V-Net | 16(20%) | 0 | 86.96 | 77.31 | 11.85 | 3.22 | 9.44 | 47.02 |
| V-Net | 80(All) | 0 | 91.62 | 84.60 | 5.40 | 1.64 | 9.44 | 47.02 |
| UA-MT (Yu et al., 2019) (MICCAI) | 8 (10%) | 72 (90%) | 86.28 | 76.11 | 18.71 | 4.63 | 9.44 | 47.02 |
| SASSNet (Li et al., 2020b) (MICCAI) | | | 85.22 | 75.09 | 11.18 | 2.89 | 9.44 | 47.05 |
| DTC (Luo et al., 2021a) (AAAI) | | | 87.51 | 78.17 | 8.23 | 2.36 | 9.44 | 47.05 |
| URPC (Luo et al., 2021b) (MICCAI) | | | 85.01 | 74.36 | 15.37 | 3.96 | 5.88 | 69.43 |
| MC-Net (Wu et al., 2021) (MICCAI) | | | 87.50 | 77.98 | 11.28 | 2.30 | 12.35 | 95.15 |
| MC-Net+ (Ours) | | | **88.96** | **80.25** | **7.93** | **1.86** | 9.44 | 47.02 |
| UA-MT (Yu et al., 2019) (MICCAI) | 16 (20%) | 64 (80%) | 88.74 | 79.94 | 8.39 | 2.32 | 9.44 | 47.02 |
| SASSNet (Li et al., 2020b) (MICCAI) | | | 89.16 | 80.60 | 8.95 | 2.26 | 9.44 | 47.05 |
| DTC (Luo et al., 2021a) (AAAI) | | | 89.52 | 81.22 | 7.07 | 1.96 | 9.44 | 47.05 |
| URPC (Luo et al., 2021b) (MICCAI) | | | 88.74 | 79.93 | 12.73 | 3.66 | 5.88 | 69.43 |
| MC-Net (Wu et al., 2021) (MICCAI) | | | 90.12 | 82.12 | 8.07 | 1.99 | 12.35 | 95.15 |
| MC-Net+ (Ours) | | | **91.07** | **83.67** | **5.84** | **1.67** | 9.44 | 47.02 |

most isolated spurious regions and preserves finer anatomical details (indicated by the purple and yellow arrows in Fig. 3) during semi-supervised left atrium segmentation.

Table 1 reports the quantitative results on the LA dataset, including the performance of a fully supervised V-Net trained with 10%, 20%, and all labeled data as reference points. By effectively exploiting the unlabeled data, MC-Net+ achieves a remarkable improvement in Dice score from 78.57% to 88.96% when only 10% of the labeled data are used for training. Moreover, with only 20% labeled data, MC-Net+ attains performance comparable to the fully supervised upper bound (*e.g.,* 91.07% vs. 91.62% in Dice). As further shown in Table 1, MC-Net+ consistently outperforms all competing methods under both semi-supervised settings while incurring no additional inference cost relative to the V-Net backbone.

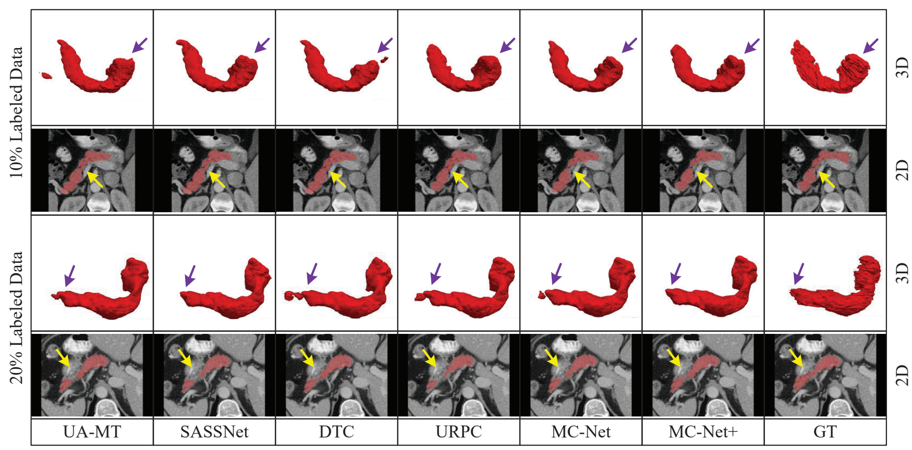


Figure 4: Exemplar segmentation results in 2D and 3D views on the Pancreas-CT dataset, produced by UA-MT (Yu et al., 2019), SASSNet (Li et al., 2020b), DTC (Luo et al., 2021a), URPC (Luo et al., 2021a), MC-Net (Wu et al., 2021), and our MC-Net+ model, with the corresponding ground truth shown on the right.

### *5.2. Performance on the Pancreas-CT Dataset*

Fig. 4 and Table 2 present the segmentation results of our model and five competing semi-supervised methods on the Pancreas-CT dataset. With the exception of the multi-scale consistency method (Luo et al., 2021b), our single-scale MC-Net+ model achieves the highest Dice and Jaccard scores among all

Table 2: Comparisons with five state-of-the-art methods on the Pancreas-CT dataset. Model complexities—the number of parameters (Para.) and multiply-accumulate operations (MACs)—are measured during inference.

| Method | # Scans used | | Metrics | | | | Complexity | |
|---|---|---|---|---|---|---|---|---|
| | Labeled | Unlabeled | Dice(%)↑ | Jaccard(%)↑ | 95HD(voxel)↓ | ASD(voxel)↓ | Para.(M) | MACs(G) |
| V-Net | 6 (10%) | 0 | 54.94 | 40.87 | 47.48 | 17.43 | 9.44 | 41.45 |
| V-Net | 12 (20%) | 0 | 71.52 | 57.68 | 18.12 | 5.41 | 9.44 | 41.45 |
| V-Net | 62 (All) | 0 | 82.60 | 70.81 | 5.61 | 1.33 | 9.44 | 41.45 |
| UA-MT (Yu et al., 2019) (MICCAI) | 6 (10%) | 56 (90%) | 66.44 | 52.02 | 17.04 | 3.03 | 9.44 | 41.45 |
| SASSNet (Li et al., 2020b) (MICCAI) | | | 68.97 | 54.29 | 18.83 | **1.96** | 9.44 | 41.48 |
| DTC (Luo et al., 2021a) (AAAI) | | | 66.58 | 51.79 | 15.46 | 4.16 | 9.44 | 41.48 |
| URPC (Luo et al., 2021b) (MICCAI) | | | **73.53** | **59.44** | 22.57 | 7.85 | 5.88 | 61.21 |
| MC-Net (Wu et al., 2021) (MICCAI) | | | 69.07 | 54.36 | **14.53** | 2.28 | 12.35 | 83.88 |
| MC-Net+ (Ours) | | | 70.00 | 55.66 | 16.03 | 3.87 | 9.44 | 41.45 |
| *Multi-scale MC-Net+** | | | ***74.01*** | ***60.02*** | ***12.59*** | *3.34* | 5.88 | 61.21 |
| UA-MT (Yu et al., 2019) (MICCAI) | 12 (20%) | 50 (80%) | 76.10 | 62.62 | 10.84 | 2.43 | 9.44 | 41.45 |
| SASSNet (Li et al., 2020b) (MICCAI) | | | 76.39 | 63.17 | 11.06 | **1.42** | 9.44 | 41.48 |
| DTC (Luo et al., 2021a) (AAAI) | | | 76.27 | 62.82 | 8.70 | 2.20 | 9.44 | 41.48 |
| URPC (Luo et al., 2021b) (MICCAI) | | | **80.02** | **67.30** | 8.51 | 1.98 | 5.88 | 61.21 |
| MC-Net (Wu et al., 2021) (MICCAI) | | | 78.17 | 65.22 | **6.90** | 1.55 | 12.35 | 83.88 |
| MC-Net+ (Ours) | | | 79.37 | 66.83 | 8.52 | 1.72 | 9.44 | 41.45 |
| *Multi-scale MC-Net+** | | | ***80.59*** | ***68.08*** | ***6.47*** | *1.74* | 5.88 | 61.21 |

* We designed our multi-scale MC-Net+ model based on (Luo et al., 2021b).

compared approaches for semi-supervised pancreas segmentation. Because the original mutual consistency constraint operates at a single scale and pancreas segmentation is a particularly challenging task that benefits from multi-scale contextual information, we further developed a multi-scale variant of MC-Net+ by incorporating the multi-scale framework of (Luo et al., 2021b). This multi-scale MC-Net+ achieves the best overall performance under both labeled-data settings on the Pancreas-CT dataset (see Table 2), demonstrating that our approach can be readily combined with multi-scale methods to further boost segmentation accuracy. It is also worth noting that our model neither relies on any post-processing modules nor employs shape-related constraints during training. Consistent with the findings on the LA dataset, our single-scale MC-Net+ model achieves competitive surface-based metrics and accurately segments challenging anatomical areas (indicated by the purple and yellow arrows in Fig. 4).

Table 3: Comparisons with five state-of-the-art methods on the ACDC dataset. Model complexities—the number of parameters (Para.) and multiply-accumulate operations (MACs)—are measured during inference.

| Method | # Scans used | | Metrics | | | | Complexity | |
|---|---|---|---|---|---|---|---|---|
| | Labeled | Unlabeled | Dice(%)↑ | Jaccard(%)↑ | 95HD(voxel)↓ | ASD(voxel)↓ | Para.(M) | MACs(G) |
| U-Net | 7 (10%) | 0 | 77.34 | 66.20 | 9.18 | 2.45 | 1.81 | 2.99 |
| U-Net | 14 (20%) | 0 | 85.15 | 75.48 | 6.20 | 2.12 | 1.81 | 2.99 |
| U-Net | 70 (All) | 0 | 91.65 | 84.93 | 1.89 | 0.56 | 1.81 | 2.99 |
| UA-MT (Yu et al., 2019) (MICCAI) | 7 (10%) | 63 (90%) | 81.58 | 70.48 | 12.35 | 3.62 | 1.81 | 2.99 |
| SASSNet (Li et al., 2020b) (MICCAI) | | | 84.14 | 74.09 | **5.03** | **1.40** | 1.81 | 3.02 |
| DTC (Luo et al., 2021a) (AAAI) | | | 82.71 | 72.14 | 11.31 | 2.99 | 1.81 | 3.02 |
| URPC (Luo et al., 2021b) (MICCAI) | | | 81.77 | 70.85 | 5.04 | 1.41 | 1.83 | 3.02 |
| MC-Net (Wu et al., 2021) (MICCAI) | | | 86.34 | 76.82 | 7.08 | 2.08 | 2.58 | 5.39 |
| MC-Net+ (Ours) | | | **87.10** | **78.06** | 6.68 | 2.00 | 1.81 | 2.99 |
| UA-MT (Yu et al., 2019) (MICCAI) | 14 (20%) | 56 (80%) | 85.87 | 76.78 | 5.06 | 1.54 | 1.81 | 2.99 |
| SASSNet (Li et al., 2020b) (MICCAI) | | | 87.04 | 78.13 | 7.84 | 2.15 | 1.81 | 3.02 |
| DTC (Luo et al., 2021a) (AAAI) | | | 86.28 | 77.03 | 6.14 | 2.11 | 1.81 | 3.02 |
| URPC (Luo et al., 2021b) (MICCAI) | | | 85.07 | 75.61 | 6.26 | 1.77 | 1.83 | 3.02 |
| MC-Net (Wu et al., 2021) (MICCAI) | | | 87.83 | 79.14 | **4.94** | **1.52** | 2.58 | 5.39 |
| MC-Net+ (Ours) | | | **88.51** | **80.19** | 5.35 | 1.54 | 1.81 | 2.99 |

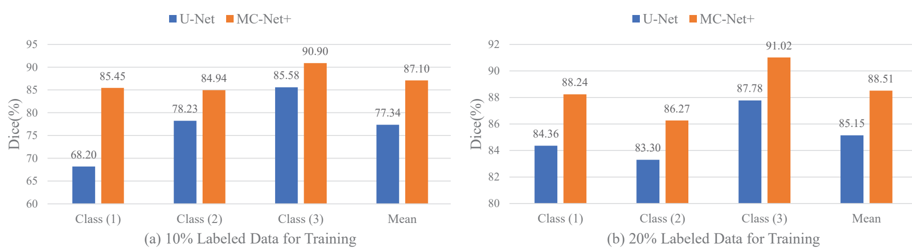


Figure 5: Per-class Dice performance on the ACDC dataset, obtained by a fully supervised U-Net model and our semi-supervised MC-Net+ model trained with 10% *(a)* and 20% *(b)* labeled data.

### *5.3. Performance on the ACDC Dataset*

We further extended our model to the 2D multi-class segmentation task. The results in Table 3 report the average performance across the three segmented structures—the myocardium, left ventricle, and right ventricle—on the ACDC dataset. The key findings are as follows: (1) compared with all competing methods, our model achieves the highest Dice and Jaccard scores, along with competitive surface-based metrics, under each semi-supervised setting; (2) by effectively leveraging the unlabeled data, MC-Net+ yields average Dice improvements of approximately 10% and 3% over the fully supervised U-Net model trained with

10% and 20% labeled data, respectively. Additionally, Fig. 5 presents the per-class Dice scores for the fully supervised U-Net model and our semi-supervised MC-Net+ model. Regardless of whether 10% or 20% labeled data are used for training, MC-Net+ delivers substantial performance gains across all classes for 2D medical image segmentation.

Taken together, the results on all three datasets demonstrate that MC-Net+ consistently outperforms other SOTA methods for semi-supervised medical image segmentation. Notably, our model is not tied to a specific backbone and can be applied to a variety of medical tasks in either 2D or 3D settings. The results also confirm that introducing an additional decoder to increase model diversity leads to improved semi-supervised segmentation performance across all datasets (*i.e.,* MC-Net+ vs. MC-Net (Wu et al., 2021)).

Furthermore, we identify three noteworthy properties of the proposed model: (1) even without shape-related constraints, the model produces segmentation results with few isolated spurious regions; (2) MC-Net+ is particularly effective at segmenting challenging anatomical structures, such as the thin branch junctions shown in Fig. 3; (3) at test time, MC-Net+ introduces no additional inference cost. These attributes make our model well suited for integration into automated CAD systems for clinical disease diagnosis.

## 6. Discussion

### *6.1. Ablation Study*

The ablation studies (Table 4) were conducted on the LA dataset to evaluate the contribution of each design component. The results reveal that, when trained with 10% or 20% labeled data: (1) the most substantial performance gains (average Dice improvements of 5.28% and 4.59%, respectively) are achieved by enforcing agreement among the three decoders (*i.e.,* reducing model uncertainty); (2) employing multiple subtly distinct decoders, denoted by $DD$, yields additional average Dice improvements of 0.63% and 0.13%. Notably, whereas the concurrent work (Chen et al., 2021) uses identical architectures with dif-

Table 4: Ablation studies of the MC-Net+ model on the LA dataset. Here, $DD$ denotes the use of different up-sampling strategies to construct three decoders, $CC$ denotes applying consistency constraints solely to either $p(y_{pred}|x;\theta_{sub})$ or $p^*(y^*_{pred}|x;\theta_{sub})$, and $MC$ denotes applying mutual consistency constraints between $p(y_{pred}|x;\theta_{sub})$ and $p^*(y^*_{pred}|x;\theta_{sub})$.

| # Scans used | | Designs | | | Metrics | | | |
|---|---|---|---|---|---|---|---|---|
| Labeled | Unlabeled | *DD* | *CC* | *MC* | Dice(%)↑ | Jaccard(%)↑ | 95HD(voxel)↓ | ASD(voxel)↓ |
| 8 (10%) | 0 | | | | 82.69 | 70.93 | 28.27 | 7.89 |
| 8 (10%) | 72 (90%) | | ✓ | | 87.61 | 78.17 | 13.65 | 3.09 |
| | | | ✓* | | 88.33 | 79.31 | 9.17 | 1.92 |
| | | | | ✓ | 88.58 | 79.68 | **7.79** | 2.01 |
| | | ✓ | ✓ | | 88.65 | 79.77 | 9.24 | 2.20 |
| | | ✓ | ✓* | | 88.70 | 79.85 | 8.08 | 2.03 |
| | | ✓ | | ✓ | **88.96** | **80.25** | 7.93 | **1.86** |
| 16 (20%) | 0 | | | | 86.01 | 75.92 | 19.27 | 4.84 |
| 16 (20%) | 64 (80%) | | ✓ | | 90.60 | 82.90 | 7.44 | 2.28 |
| | | | ✓* | | 90.60 | 82.91 | 6.13 | 1.88 |
| | | | | ✓ | 90.84 | 83.32 | 5.89 | 1.85 |
| | | ✓ | ✓ | | 90.77 | 83.20 | 8.27 | 2.50 |
| | | ✓ | ✓* | | 90.63 | 83.03 | 5.99 | **1.61** |
| | | ✓ | | ✓ | **91.07** | **83.67** | **5.84** | 1.67 |
| 80 (100%) | 0 | | | | 92.05 | 85.33 | 7.10 | 1.79 |

* The consistency constraints are enforced on $p^*(y^*_{pred}|x;\theta_{sub})$ of different sub-models.

ferent initialization parameters, our approach leverages different up-sampling strategies to enhance intra-model diversity, resulting in superior performance; (3) the mutual consistency training scheme, denoted by $MC$, consistently outperforms the application of consistency constraints to either probability outputs or soft pseudo labels alone (denoted by $CC$ or $CC^*$, respectively).

We also include the fully supervised MC-Net+ model (*i.e.,* trained without $L_{mc}$) as a reference baseline. The results demonstrate that merely adopting three subtly distinct decoders does not yield substantial performance gains; it is the enforcement of our mutual consistency constraints that significantly improves semi-supervised segmentation performance, with overall Dice improvements of 6.25% and 5.07% under the two labeled-data settings, respectively.

### 6.2. Effects of Different Up-sampling Strategies

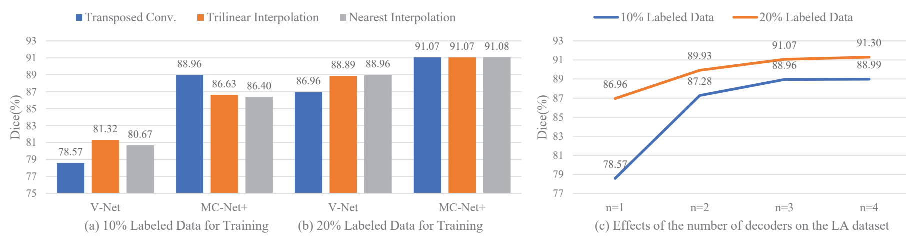


Figure 6: Dice performance with different up-sampling strategies *(a* and *b)* and varying numbers of decoders *(c)* on the LA dataset. Note that, to construct four decoders, we employ two transposed convolutional layers with different initialization parameters owing to the limited number of available up-sampling strategies.

To increase intra-model diversity, MC-Net+ adopts a transposed convolutional layer, a bilinear interpolation layer, and a nearest-neighbor interpolation layer to construct three subtly distinct decoders. Fig. 6 shows that the Dice scores of fully supervised V-Net models vary considerably across different up-sampling strategies on the LA dataset. However, when our mutual consistency constraints are applied, MC-Net+ achieves consistently improved results, and the three decoders converge toward invariant outputs, yielding fewer ambiguous predictions and lower model uncertainty. Because all three decoders produce similar results, we select only the original encoder–decoder architecture—*i.e.,* the shared encoder and the first decoder—as the final test-time model to minimize inference cost. It is worth noting that our earlier MC-Net model (Wu et al., 2021) averaged the outputs of two decoders as the final prediction, whereas the new MC-Net+ model uses only the first decoder's output during inference.

Because the number of decoders $n$ is a flexible design choice, we further conducted a sensitivity analysis to examine its effect. Fig. 6 (c) shows that increasing the number of decoders can improve performance, but the marginal gains diminish due to confirmation bias (Li et al., 2020a). When labeled data are extremely scarce, deep models may generate incorrect predictions with unwarranted high confidence. We therefore set $n = 3$ in this work to achieve a balance between effectiveness and efficiency. We conjecture that with more abundant labeled data, a model with additional diverse sub-models could achieve even better segmentation performance.

### 6.3. *Effects of Temperature $T$*

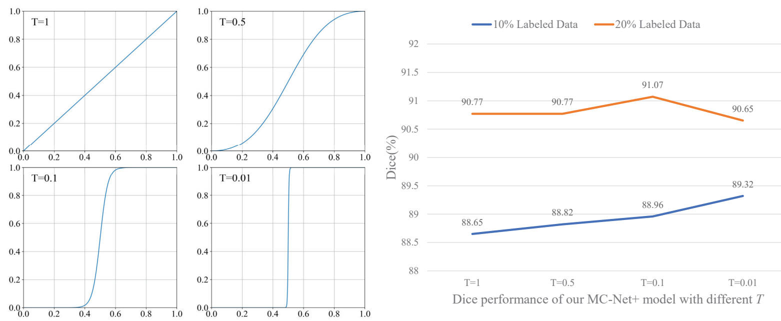


Figure 7: Illustration of the sharpening function with different temperatures *(left)* and the corresponding Dice performance of MC-Net+ on the LA dataset *(right)*.

To mitigate the influence of easily misclassified pixels, MC-Net+ employs soft pseudo labels to impose the entropy-minimization constraint while avoiding excessive sharpening of predictions near the probability value of 0.5 (see the left panel of Fig. 7). The right panel of Fig. 7 presents the Dice performance of MC-Net+ trained with different temperatures $T$ on the LA dataset. The Dice scores remain relatively stable across different values of $T$ under each semi-supervised setting, indicating that our model is fairly robust to this hyper-parameter. A larger $T$ provides insufficient entropy minimization, whereas a smaller $T$ amplifies the noise in the pseudo labels, potentially leading to error accumulation. Accordingly, we adopt a temperature of $T = 0.1$ for generating soft pseudo labels on all datasets.

### 6.4. *Effects of Loss Weight $\lambda$*

We further investigated the sensitivity of MC-Net+ to the loss weight $\lambda$, which balances the supervised loss and the mutual consistency loss, on the LA and Pancreas-CT datasets (see Fig. 8). A smaller $\lambda$ tends to degrade performance because the limited labeled data provide insufficient supervision for

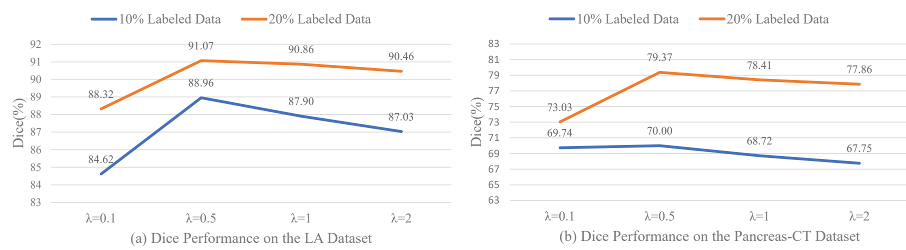


Figure 8: Dice performance with different loss weights $\lambda$ on the LA and Pancreas-CT datasets.

accurate decoder outputs, whereas a larger $\lambda$ weakens the mutual consistency constraint, also resulting in suboptimal performance. Based on this analysis, we set $\lambda = 0.5$ for the 3D segmentation tasks on both the LA and Pancreas-CT datasets.

### *6.5. Effects of Different Distance Measurements D*

Table 5: Comparison of different distance measurements $D$ on the LA dataset.

| Method | Output | # Scans used | | Metrics | | | |
|---|---|---|---|---|---|---|---|
| | | Labeled | Unlabeled | Dice(%)↑ | Jaccard(%)↑ | 95HD(voxel)↓ | ASD(voxel)↓ |
| MC-Net+ w/ KL | First Decoder | 8 (10%) | 72 (90%) | 87.86 | 78.86 | 9.65 | 2.27 |
| | Mean | | | 88.04 | 79.08 | 9.40 | 2.32 |
| MC-Net+ w/ MSE | First Decoder | 8 (10%) | 72 (90%) | **88.96** | **80.25** | **7.93** | **1.86** |
| | Mean | | | **88.94** | **80.22** | **8.02** | **1.83** |
| MC-Net+ w/ KL | First Decoder | 16 (20%) | 64 (80%) | 90.93 | 83.45 | 6.04 | **1.63** |
| | Mean | | | 90.96 | 83.50 | 6.08 | **1.60** |
| MC-Net+ w/ MSE | First Decoder | 16 (20%) | 64 (80%) | **91.07** | **83.67** | **5.84** | 1.67 |
| | Mean | | | **91.08** | **83.70** | **5.90** | **1.60** |

We also investigated the impact of employing different distance measures $D$ for quantifying the discrepancy among multiple outputs on the LA dataset. Table 5 reports the results of MC-Net+ when trained with the Kullback–Leibler (KL) divergence. Although the KL-based loss also improves segmentation performance under each semi-supervised setting, the simpler MSE loss proves sufficient to demonstrate the effectiveness of our model. We therefore adopt MSE as the distance measure $D$ throughout this work.

### 6.6. Limitations and Future Work

Although MC-Net+ is both simple and effective for semi-supervised medical image segmentation, its design still requires multiple pre-defined decoders, and the set of available up-sampling strategies is inherently limited. For novel tasks, exploring more diverse model architectures to further increase intra-model diversity remains a worthwhile direction. Moreover, this work considers only model-level perturbations (*i.e.,* different up-sampling strategies), whereas data-level perturbations may also prove beneficial. However, certain data-agnostic augmentation operations such as ColorJitter (Sohn et al., 2020) may not be well suited for medical imaging data. Future work will focus on developing medical-data-specific perturbation strategies and evaluating the proposed model on larger-scale datasets.

## 7. Conclusion

In this paper, we have presented MC-Net+, a novel mutual consistency network for semi-supervised medical image segmentation. Effective exploitation of challenging regions plays a pivotal role in semi-supervised segmentation. The architectural design employing three subtly distinct decoders identifies highly uncertain areas, while the proposed mutual consistency constraint between probability outputs and soft pseudo labels establishes an end-to-end training framework that compels the model to produce consistent, low-entropy predictions in difficult regions. Extensive experiments on three medical datasets demonstrate that MC-Net+ achieves superior performance over five existing methods, establishing a new state of the art for semi-supervised medical image segmentation.

## References

Abdar, M., Pourpanah, F., Hussain, S., Rezazadegan, D., Liu, L., Ghavamzadeh, M., Fieguth, P., Cao, X., Khosravi, A., Acharya, U.R., et al., 2021. A review of uncertainty quantification in deep learning: Techniques, applications and challenges. Information Fusion .

Bernard, O., Lalande, A., Zotti, C., Cervenansky, F., Yang, X., Heng, P.A., Cetin, I., Lekadir, K., Camara, O., Ballester, M.A.G., et al., 2018. Deep learning techniques for automatic mri cardiac multi-structures segmentation and diagnosis: Is the problem solved? IEEE transactions on medical imaging 37, 2514–2525.

Castillo-Navarro, J., Le Saux, B., Boulch, A., Lefèvre, S., 2020. On auxiliary losses for semi-supervised semantic segmentation, in: Proceedings of the European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases, pp. 1–10.

Chaitanya, K., Erdil, E., Karani, N., Konukoglu, E., 2020. Contrastive learning of global and local features for medical image segmentation with limited annotations. Advances in Neural Information Processing Systems 33, 12546–12558.

Chen, X., Yuan, Y., Zeng, G., Wang, J., 2021. Semi-supervised semantic segmentation with cross pseudo supervision, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 2613–2622.

Clark, K., Vendt, B., Smith, K., Freymann, J., Kirby, J., Koppel, P., Moore, S., Phillips, S., Maffitt, D., Pringle, M., et al., 2013. The cancer imaging archive (tcia): maintaining and operating a public information repository. Journal of digital imaging 26, 1045–1057.

Esser, P., Sutter, E., Ommer, B., 2018. A variational u-net for conditional appearance and shape generation, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 8857–8866.

Gal, Y., Ghahramani, Z., 2016. Dropout as a bayesian approximation: Representing model uncertainty in deep learning, in: International Conference on Machine Learning, PMLR. pp. 1050–1059.

Hesamian, M.H., Jia, W., He, X., Kennedy, P., 2019. Deep learning techniques for medical image segmentation: achievements and challenges. Journal of digital imaging 32, 582–596.

Jin, L., Lu, H., Wen, G., 2019. Fast uncertainty quantification of reservoir simulation with variational u-net. arXiv preprint arXiv:1907.00718 .

Jungo, A., Reyes, M., 2019. Assessing reliability and challenges of uncertainty estimations for medical image segmentation, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer. pp. 48–56.

Kalluri, T., Varma, G., Chandraker, M., Jawahar, C., 2019. Universal semi-supervised semantic segmentation, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 5259–5270.

Kendall, A., Gal, Y., 2017. What uncertainties do we need in bayesian deep learning for computer vision? arXiv preprint arXiv:1703.04977 .

Laine, S., Aila, T., 2016. Temporal ensembling for semi-supervised learning. arXiv preprint arXiv:1610.02242 .

Lakshminarayanan, B., Pritzel, A., Blundell, C., 2016. Simple and scalable predictive uncertainty estimation using deep ensembles. arXiv preprint arXiv:1612.01474 .

Lee, D.H., et al., 2013. Pseudo-label: The simple and efficient semi-supervised learning method for deep neural networks, in: Workshop on challenges in representation learning, ICML, p. 896.

Li, J., Socher, R., Hoi, S.C., 2020a. Dividemix: Learning with noisy labels as semi-supervised learning. arXiv preprint arXiv:2002.07394 .

Li, S., Zhang, C., He, X., 2020b. Shape-aware semi-supervised 3d semantic segmentation for medical images, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer. pp. 552–561.

Luo, X., 2020. SSL4MIS. https://github.com/HiLab-git/SSL4MIS.

Luo, X., Chen, J., Song, T., Wang, G., 2021a. Semi-supervised medical image segmentation through dual-task consistency, in: Proceedings of the AAAI Conference on Artificial Intelligence, pp. 8801–8809.

Luo, X., Liao, W., Chen, J., Song, T., Chen, Y., Zhang, S., Chen, N., Wang, G., Zhang, S., 2021b. Efficient semi-supervised gross target volume of nasopharyngeal carcinoma segmentation via uncertainty rectified pyramid consistency, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer. pp. 318–329.

Ma, J., Wei, Z., Zhang, Y., Wang, Y., Lv, R., Zhu, C., Gaoxiang, C., Liu, J., Peng, C., Wang, L., et al., 2020. How distance transform maps boost segmentation cnns: an empirical study, in: Medical Imaging with Deep Learning, PMLR. pp. 479–492.

Milletari, F., Navab, N., Ahmadi, S.A., 2016. V-net: Fully convolutional neural networks for volumetric medical image segmentation, in: 2016 Fourth International Conference on 3D Vision (3DV), IEEE. pp. 565–571.

Mittal, S., Tatarchenko, M., Brox, T., 2019. Semi-supervised semantic segmentation with high-and low-level consistency. IEEE Transactions on Pattern Analysis and Machine Intelligence 43, 1369–1379.

Miyato, T., Maeda, S.i., Koyama, M., Ishii, S., 2018. Virtual adversarial training: a regularization method for supervised and semi-supervised learning. IEEE Transactions on Pattern Analysis and Machine Intelligence 41, 1979–1993.

Murugesan, B., Sarveswaran, K., Shankaranarayana, S.M., Ram, K., Joseph, J., Sivaprakasam, M., 2019. Psi-net: Shape and boundary aware joint multi-task deep network for medical image segmentation, in: Proceedings of the Annual International Conference of the IEEE Engineering in Medicine and Biology Society, IEEE. pp. 7223–7226.

Ouali, Y., Hudelot, C., Tami, M., 2020. Semi-supervised semantic segmentation with cross-consistency training, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 12674–12684.

Pham, H., Dai, Z., Xie, Q., Le, Q.V., 2021. Meta pseudo labels, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 11557–11568.

Qiao, F., Peng, X., 2021. Uncertainty-guided model generalization to unseen domains, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 6790–6800.

Rizve, M.N., Duarte, K., Rawat, Y.S., Shah, M., 2021. In defense of pseudo-labeling: An uncertainty-aware pseudo-label selection framework for semi-supervised learning. arXiv preprint arXiv:2101.06329 .

Ronneberger, O., Fischer, P., Brox, T., 2015. U-net: Convolutional networks for biomedical image segmentation, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, pp. 234–241.

Sohn, K., Berthelot, D., Li, C.L., Zhang, Z., Carlini, N., Cubuk, E.D., Kurakin, A., Zhang, H., Raffel, C., 2020. Fixmatch: Simplifying semi-supervised learning with consistency and confidence. arXiv preprint arXiv:2001.07685 .

Soviany, P., Ionescu, R.T., Rota, P., Sebe, N., 2021. Curriculum learning: A survey. arXiv preprint arXiv:2101.10382 .

Tarvainen, A., Valpola, H., 2017. Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results. arXiv preprint arXiv:1703.01780 .

Wang, K., Zhan, B., Zu, C., Wu, X., Zhou, J., Zhou, L., Wang, Y., 2021. Tripled-uncertainty guided mean teacher model for semi-supervised medical image segmentation, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer. pp. 450–460.

Wang, Y., Huang, G., Song, S., Pan, X., Xia, Y., Wu, C., 2022. Regularizing deep networks with semantic data augmentation. IEEE Transactions on Pattern Analysis and Machine Intelligence 44, 3733–3748.

Wu, Y., Wu, Z., Wu, Q., Ge, Z., Cai, J., 2022. Exploring smoothness and class-separation for semi-supervised medical image segmentation. arXiv preprint arXiv:2203.01324 .

Wu, Y., Xu, M., Ge, Z., Cai, J., Zhang, L., 2021. Semi-supervised left atrium segmentation with mutual consistency training, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer. pp. 297–306.

Xia, Y., Liu, F., Yang, D., Cai, J., Yu, L., Zhu, Z., Xu, D., Yuille, A., Roth, H., 2020a. 3d semi-supervised learning with uncertainty-aware multi-view co-training, in: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, pp. 3646–3655.

Xia, Y., Yang, D., Yu, Z., Liu, F., Cai, J., Yu, L., Zhu, Z., Xu, D., Yuille, A., Roth, H., 2020b. Uncertainty-aware multi-view co-training for semi-supervised medical image segmentation and domain adaptation. Medical Image Analysis 65, 101766.

Xie, Q., Dai, Z., Hovy, E., Luong, M.T., Le, Q.V., 2019. Unsupervised data augmentation for consistency training. arXiv preprint arXiv:1904.12848 .

Xie, Y., Zhang, J., Liao, Z., Verjans, J., Shen, C., Xia, Y., 2020. Pairwise relation learning for semi-supervised gland segmentation, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer. pp. 417–427.

Xiong, Z., Xia, Q., Hu, Z., Huang, N., Bian, C., Zheng, Y., Vesal, S., Ravikumar, N., Maier, A., Yang, X., et al., 2021. A global benchmark of algorithms for segmenting the left atrium from late gadolinium-enhanced cardiac magnetic resonance imaging. Medical Image Analysis 67, 101832.

You, C., Dai, W., Staib, L., Duncan, J.S., 2022a. Bootstrapping semi-supervised medical image segmentation with anatomical-aware contrastive distillation. arXiv preprint arXiv:2206.02307 .

You, C., Yang, J., Chapiro, J., Duncan, J.S., 2020. Unsupervised wasserstein distance guided domain adaptation for 3d multi-domain liver segmentation, in: Interpretable and Annotation-Efficient Learning for Medical Image Computing. Springer, pp. 155–163.

You, C., Zhao, R., Liu, F., Chinchali, S., Topcu, U., Staib, L., Duncan, J.S., 2022b. Class-aware generative adversarial transformers for medical image segmentation. arXiv preprint arXiv:2201.10737 .

You, C., Zhao, R., Staib, L., Duncan, J.S., 2021. Momentum contrastive voxel-wise representation learning for semi-supervised volumetric medical image segmentation. arXiv preprint arXiv:2105.07059 .

You, C., Zhou, Y., Zhao, R., Staib, L., Duncan, J.S., 2022c. Simcvd: Simple contrastive voxel-wise representation distillation for semi-supervised medical image segmentation. IEEE Transactions on Medical Imaging .

Yu, L., Wang, S., Li, X., Fu, C.W., Heng, P.A., 2019. Uncertainty-aware self-ensembling model for semi-supervised 3d left atrium segmentation, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer. pp. 605–613.

Zhang, H., Cisse, M., Dauphin, Y.N., Lopez-Paz, D., 2017. mixup: Beyond empirical risk minimization. arXiv preprint arXiv:1710.09412 .

Zhang, Y., Xiang, T., Hospedales, T.M., Lu, H., 2018. Deep mutual learning, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 4320–4328.

Zheng, Z., Yang, Y., 2021. Rectifying pseudo label learning via uncertainty estimation for domain adaptive semantic segmentation. International Journal of Computer Vision 129, 1106–1120.

Zhou, Y., Li, Z., Bai, S., Wang, C., Chen, X., Han, M., Fishman, E., Yuille, A.L., 2019. Prior-aware neural network for partially-supervised multi-organ segmentation, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 10672–10681.

Zhu, J., Li, Y., Hu, Y., Ma, K., Zhou, S.K., Zheng, Y., 2020. Rubik's cube+: A self-supervised feature learning framework for 3d medical image analysis. Medical image analysis 64, 101746.